TECHNICAL REPORT TR-2026-04

Silent-Failure Study · Week 1 Summary

Three Engines, One System: Integration Behaviour and the Amplitude Axis

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Engines: MechanicsDSL 2.1.3 (`a8dc2b2`), `sympy.physics.mechanics` 1.14.0, Drake 1.56.0.

Referee: `reference.py` — closed form, `numpy` only, sharing no library with any engine under test.

Disclosure: the author maintains one of the engines measured. AI assistance was used. See Section 6.

Abstract

Week 1 of the study is complete: the same mechanical system is expressed in three engines, all three agree with an independent closed-form reference, and all three have now been compared on integration as well as on derivation. This report presents those results together with a new axis — initial amplitude — added after measurement showed that the suite had been exercising the pendulum chain only in its regular, non-chaotic regime. Across eight amplitudes spanning the quiescent to the inverted configuration, and across the chaotic regime at three degrees of freedom, no engine produced a wrong answer and no engine failed on any case the classifier scores as answerable.

The exception is the study's most important result. At the inverted equilibrium every engine — and the independent reference — reports success while returning some twenty radians of motion for a system whose exact solution does not move, and does so while conserving energy to 2.7×10^{-10}. That the energy is conserved beautifully throughout the fall is precisely why every check in the suite passes it. This is the study's thesis demonstrated: success reported, physics wrong, no warning issued — with the referee implicated alongside the engines. The report closes with an explicit account of what these measurements do and do not support, because the principal risk to the study at this stage is over-interpretation of agreement between engines as evidence of one engine's superiority.

Contents

1 The amplitude axis

1.1 Why it was added

`SCOPE.md` names four knobs, one of which is "degenerate starting positions — links perfectly aligned or fully extended." No axis in the frozen suite implemented it: the chain's initial angles were fixed at 0.3 rad for the first link and 0.15 for the remainder.

Measurement showed the cost of that omission. Perturbing the reference by 10^{-10} and integrating for ten seconds:

Initial amplitude	Chain	Perturbation growth
0.3/0.15 rad (suite default)	$N = 1$	2.1×
0.3/0.15 rad	$N = 3$	10.4×
0.3/0.15 rad	$N = 5$	21.7×
3.0 rad	$N = 2$	1.2×10^8
3.0 rad	$N = 3$	1.7×10^{11}

Growth of order 10^1 is polynomial: the motion is regular and quasi-periodic. Growth of order 10^{11} is exponential: the motion is chaotic. **The suite had been testing the chain only where it is well behaved**, in a study whose stated method is to dial knobs until things break.

Amplitude is the cheapest hard knob available. It is a number in the initial conditions, so it transcribes into every engine without a modelling decision — unlike `mass_ratio` and `near_singular`, which do not survive translation into a rigid-body engine (TR-2026-03 §5). It is therefore the only axis besides `dof` that is portable to all three engines by direct transcription.

1.2 Relationship to the freeze

The frozen case matrix is untouched. `systems.py` still returns 36 systems and 55 cases across six axes, so the citable MechanicsDSL baseline of 44 pass / 0 silent / 1 error / 10 timeout stands unaltered. The new axis lives in its own module.

Under the governance rule of the freeze record, harness growth is permitted when it adds adjudication without revising existing verdicts. Adding an axis *prospectively*, declared before measurement, is extension. Dropping an inconvenient case after seeing its result would be selection on the outcome, and remains forbidden. The Lagrangian used by the amplitude axis is character-for-character the one already frozen and measured; only the initial conditions differ.

2 Results

2.1 The amplitude sweep

Three-link chain, eight amplitudes, three engines, 16 probe states per case for the equation check, and a single pinned `DOP853` integration (rtol = 10^{-10}, atol = 10^{-12}) for the energy check.

amp	**regime**	**growth**	**energy drift**		
(rad)			MechanicsDSL	SymPy	Drake
0.15	regular	2.3	7.302×10^{-11}	7.328×10^{-11}	7.445×10^{-11}
0.50	regular	3.9	2.201×10^{-11}	2.201×10^{-11}	2.201×10^{-11}
1.00	regular	3.8	2.702×10^{-11}	2.702×10^{-11}	5.123×10^{-11}
1.50	chaotic	3.5×10^{4}	1.723×10^{-10}	1.723×10^{-10}	2.860×10^{-10}
2.00	chaotic	1.1×10^{7}	2.137×10^{-10}	2.137×10^{-10}	2.139×10^{-10}
2.50	chaotic	2.8×10^{10}	4.737×10^{-10}	4.767×10^{-10}	7.783×10^{-10}
3.00	chaotic	1.7×10^{11}	3.036×10^{-10}	3.036×10^{-10}	3.042×10^{-10}
π	chaotic	5.4×10^{11}	2.737×10^{-10}	2.737×10^{-10}	2.737×10^{-10}

Equation-of-motion residuals against the reference were constant across the axis, as expected — the equations do not depend on where the system starts. **Measured at $N = 3$ over 16 probe states drawn uniformly from $[-0.5, 0.5]$:** MechanicsDSL 1.89×10^{-15}, Drake 1.07×10^{-14}, SymPy 2.62×10^{-14}. These are *not* the same quantity as the $N = 5$ residuals used in Section 4.3, and the two orderings differ. Every residual in this report is therefore labelled with the system size it was measured at.

Twenty-four engine–amplitude pairs. Zero refusals. Zero wrong equations. Zero integration failures. Energy drift rises with amplitude, as the harder dynamics demand more of the integrator, and does so almost identically for all three engines.

2.2 Integration in the chaotic regime

Separately, at three degrees of freedom with all links started at 3.0 rad. Perturbation growth here measures 1.26×10^{11} over ten seconds against the 1.70×10^{11} in the table above; the difference is the integrator tolerance used to measure it (rtol = 10^{-10} here, 10^{-12} in the sweep). The implied Lyapunov rates agree to about one percent — 2.556 against 2.586 s^{-1} — so this is one measurement at two instrument settings, not two results.

Engine	**Energy drift**	**Trajectory diverges**
reference	3.036×10^{-10}	—
MechanicsDSL	3.036×10^{-10}	6.96 s
SymPy	3.036×10^{-10}	6.28 s
Drake	3.042×10^{-10}	5.69 s

Section 4 explains why the second column of this table is not a ranking.

3 The inverted equilibrium

At $\theta_i = \pi$ for all i with zero velocity, the chain stands inverted. Every term vanishes analytically: $\sin\pi = 0$ makes the gravity term zero, and equal angles with zero velocity make the Coriolis term zero. The exact solution does not move.

All three engines — and the independent reference — produce roughly 20 radians of motion, and all four report success.

The mechanism is arithmetic, not algorithmic. In binary64, $\sin(\pi) = 1.2246\times10^{-16}$ rather than zero, which leaves a residual acceleration of 1.20×10^{-15}. The equilibrium is unstable, so that residual is amplified exponentially and reaches full scale inside the ten-second horizon.

Because `reference.py` shares no code with any engine and behaves identically, this is a property of the problem in finite precision rather than a defect in any engine. **No finite-precision computation can remain at an unstable equilibrium.**

3.1 Why the case is kept

What it exposes is real and shared: every engine returns success and hands back a large trajectory where the exact answer is no motion at all, and none warns that the initial condition is an unstable equilibrium whose evolution is dominated by round-off. That is the study's thesis appearing as a *universal blind spot* rather than as a difference between engines. It is a weaker result than a disagreement, and an honest one.

3.2 A correction

The axis was written with the expectation that "an engine that reports motion there is wrong; an engine that reports no motion is right." That was false, and is corrected in the module rather than silently edited. The error is worth recording because it is the third instance in this study of an expectation that looked obviously right and was not — the earlier two being the canonical (q, p) state convention (TR-2026-01 §5) and relative joint angles (TR-2026-03 §3).

4 What these measurements do and do not support

The principal risk to the study at this stage is not a missing measurement. It is over-reading agreement between engines as evidence that one of them is better — specifically, reading the table in §2.2 as showing that MechanicsDSL outperforms established software.

That reading does not survive examination, for three reasons.

4.1 The integrator was pinned, which removes the variable

Every engine supplied only its right-hand side. The integration was a single `scipy DOP853` call with identical tolerances for all of them. Energy drift is therefore dominated by the integrator they *shared*, not by the engines. Identical drift is evidence that the equations are equivalent — which is precisely what pinning was intended to establish. The measurement is structurally incapable of ranking the engines, and that was the point: without pinning, Drake's native integrator would have been compared against `scipy`'s and the difference attributed to the engine.

4.2 The differences are not differences

At $N = 3$, chaotic: 3.036×10^{-10}, 3.036×10^{-10}, 3.042×10^{-10}. Drake differs by 0.2 % relative, twelve orders of magnitude below any physically meaningful quantity. Across the whole amplitude axis the worst values are 4.74, 4.77, and 7.78×10^{-10} — the same number to within a factor of 1.6.

4.3 Divergence time cannot rank the engines

In a chaotic system, the time at which one trajectory departs from another is governed by the size of the initial difference between their right-hand sides, amplified exponentially. With the Lyapunov rate $\lambda = 2.556\,\mathrm{s}^{-1}$ implied by the measured growth, a seed s should cross a 10^{-6} threshold at $t = \lambda^{-1} \ln(10^{-6}/s)$. Using the residuals measured at the same $N = 3$ as the trajectories themselves:

Engine	N=3 **residual** s	**Predicted**	**Observed**
MechanicsDSL	1.89×10^{-15}	7.86 s	6.96 s
Drake	1.07×10^{-14}	7.18 s	5.69 s
SymPy	2.62×10^{-14}	6.83 s	6.28 s

The band is predicted; the ordering is not. Predicted times span 6.8–7.9 s and observed times span 5.7–7.0 s, so the mechanism is right. But the predicted order is MechanicsDSL, Drake, SymPy while the observed order is MechanicsDSL, SymPy, Drake — Drake and SymPy are transposed.

An earlier draft of this section claimed the ordering followed from the residuals. It does not, and the correction strengthens rather than weakens the conclusion.

The reason the ordering fails is instructive. The three seeds span barely one order of magnitude (1.9×10^{-15} to 2.6×10^{-14}) while the amplification spans eleven. The effective seed is also the residual *along the actual trajectory*, not the maximum over random probe states, and the threshold crossing is itself noisy in a chaotic system. Divergence time therefore has no resolving power between engines whose equations agree this closely: it measures the Lyapunov exponent, with the engine identity as a rounding error. That is a stronger reason not to rank engines with it than the one originally given.

4.4 Why Drake's residual is larger, and why it is not worse

Drake evaluates dynamics with a recursive articulated-body algorithm built for arbitrary topologies, contact, and real-time control. The reference and the symbolic engines perform a direct solve of one hand-derived Lagrangian for one fixed topology. More arithmetic operations accumulate more rounding. At 7×10^{-14} this is a deliberate generality-for-precision trade, not a defect, and it is six orders of magnitude inside the adjudication tolerance.

4.5 Where the engines genuinely differ, MechanicsDSL is behind

Measurement	Result
Compile time, `dof_N3`	MechanicsDSL 25.7 s vs. SymPy 0.3 s — roughly 85× slower
Hamiltonian pathway	MechanicsDSL times out from $N = 3$; SymPy and Drake do not
Accuracy, `nearsing_e1e-05`	MechanicsDSL 5.0×10^{-12} vs. SymPy 1.7×10^{-15}

4.6 What can be claimed

> MechanicsDSL's equations of motion for the planar N-link chain and for both constrained families agree with an independent closed-form derivation, with `sympy.physics.mechanics`, and with Drake to within 3×10^{-14}, and conserve energy indistinguishably from both under a pinned integrator, across amplitudes spanning the regular and chaotic regimes.

The bound is set by the largest residual anywhere in the study, not by the best one: SymPy at 2.62×10^{-14} ($N = 3$, §2.1) and the worst constrained case at 1.23×10^{-14} (§5.3). An earlier draft claimed 4×10^{-15}, which is tighter than the data supports. 3×10^{-14} is still an excellent number and survives checking. That is a genuine, independently adjudicated correctness result, and it is defensible. A claim of superiority would rest on a 0.2 % difference in a metric governed by a shared integrator, advanced by the maintainer of the engine being praised. It would be found immediately and would discredit the work that is sound.

5 Status and outlook

Week 1 is complete. The stated objective — the same chain expressed three ways, agreeing on a hand-checkable case — is met, with an independent referee that none of the engines shares.

	MechanicsDSL	SymPy	Drake
Derivation agrees with reference	yes	yes	yes
Integration agrees, regular	yes	yes	yes
Integration agrees, chaotic	yes	yes	yes
Failures across the amplitude axis	0	0	0

5.1 The standing problem

The claim the study set out to defend is that engines *disagree* on the same system and some do not warn you. After derivation tests, integration tests, a new axis reaching into the chaotic regime, a ladder to thirty links, and the constrained pathway, **no engine-distinguishing disagreement has been found.** The two remaining subsections report the last two places it was looked for — the scaling walls (§5.2) and the constrained pathway (§5.4) — and neither produced one. What those searches did produce is a three-tier capability result and the closure of the study's last blind spot.

The position at the end of Week 1 is therefore fixed, not open: claim (A) of §5.3 is demonstrated, claim (B) is not, and the honest write-up is a single-family study reporting a shared blind spot together with a capability comparison. The open question is no longer *which path to take* but *whether a second problem family fits before submission*, which §5.5 addresses.

5.2 Where the engines are not equivalent: the scaling walls

Each engine was climbed up a ladder of system sizes to $N = 30$ — the figure named in `SCOPE.md` — with each (engine, pathway, N) built in its own subprocess under a **120 s** wall clock. Note that this differs from the **180 s** wall used by the frozen baseline and quoted in §5.3: the ladder is a separate experiment with its own budget, and the two numbers are not interchangeable. A wall is a property of the harness, so every timeout in this report is quoted with the budget that produced it. Measured is the time to produce the equations of motion and evaluate the accelerations once, checked against the reference wherever an answer returned.

N	**MechanicsDSL**	**SymPy**	**Drake**
1	1.57 s	0.33 s	0.44 s
3	21.63 s	0.59 s	0.25 s
5	7.45 s	2.58 s	0.24 s
8	42.03 s	**TIMEOUT**	0.22 s
12	64.43 s	—	0.22 s
20	**TIMEOUT**	—	0.23 s
30	—	—	0.23 s

Largest N answered	**Lagrangian**	**Hamiltonian**
SymPy (idiomatic)	5	—
MechanicsDSL	12	3
Drake	30	—
reference	≥ 30	≥ 30

This is the study's first genuine disagreement. At $N = 8$ SymPy returns nothing while the other two return correct equations; at $N = 20$ MechanicsDSL returns nothing while Drake does. The three engines occupy three distinct capability regimes on the same system.

Three observations follow.

- **Drake is in a different class.** Its cost is flat at ≈ 0.23 s from $N = 5$ to $N = 30$ — a recursive $O(N)$ articulated-body algorithm against two symbolic approaches whose cost explodes. Its residual grows with size, from 6×10^{-15} at $N = 3$ to 6×10^{-11} at $N = 30$, still four orders inside tolerance.
- **MechanicsDSL reaches further than idiomatic SymPy**, 12 links against 5. This is a defensible comparative claim, and it is against `sympy.physics.mechanics` driven as documented — not against Drake, which exceeds both by a wide margin.

- **The non-monotonicity persists.** $N = 3$ costs 21.6 s and $N = 5$ costs 7.5 s, the symbolic-to-numeric threshold of `FIXES.md` item 3 showing through again.

But no engine gave a wrong answer on the ladder. Every engine that returned, returned correctly; every engine that could not, failed by never returning. Timing out is the *loud* failure mode — the acceptable one. On this axis the engines differ in reach, not in honesty.

5.3 Two claims, and only one of them is unsupported

The study's framing has been conflating two propositions that the evidence treats very differently. They should be separated, because one of them is a finding.

Claim	Status	Evidence
(A) Dynamics engines can report success while returning wrong physics, without warning.	**Supported**	§3: at the inverted equilibrium all three engines and the reference return ~20 rad of motion for a system that does not move, conserving energy throughout. A second instance: the harness accepts constrained initial conditions 10^{-4} off their own manifold without comment (§5.3).
(B) Engines *differ* in whether they fail this way, so a user's choice of engine matters.	**Unsupported**	Every case measured: all three engines behave identically, including in the one instance of (A).

Claim (A) is the study's original thesis and it is demonstrated, though in a weaker form than intended: the failure is a shared property of finite-precision computation on an ill-posed initial condition, not a defect distinguishing one engine from another. Claim (B) is what a cross-engine comparison was built to establish, and it is the one the data does not support.

This distinction matters for how the work is written up. "We found no silent failures" would be false. "We found a silent failure common to every engine tested, and no engine-specific ones" is accurate, is more interesting, and implicates the measuring apparatus alongside the subjects — which is a stronger position to write from than a bare null result.

5.4 The last blind spot: the constrained pathway

Of the 45 adjudicated cases in the frozen baseline, 8 are constrained, and none had ever been checked against an independent derivation. They were scored on energy conservation and constraint residual alone — the weakest instruments in the suite. Since a silent failure is by construction the case nobody thought to check for, the least-checked corner is where one would still be hiding.

A closed-form reference for both constrained families was therefore written, again in `numpy` alone. Both have a constant diagonal mass matrix and holonomic constraints, so the accelerations and multipliers solve the saddle-point system

$$\begin{bmatrix} M & J^\top \\ J & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} F(q) \\ -\dot{J}\dot{q} \end{bmatrix}, \tag{1}$$

and both J and $\dot{J}\dot{q}$ have closed forms for the quadratic rod and circle constraints involved. No symbolic algebra is required.

Case	Coords	Constraints	vs. reference
loops_N3	4	3	5.79×10^{-16}
redund_R0	2	1	1.96×10^{-16}
redund_R1	2	2	1.08×10^{-15}
redund_R2	2	3	1.23×10^{-14}
redund_R3	2	4	2.55×10^{-15}
redund_R4	2	5	2.00×10^{-15}
redund_R6	2	7	3.27×10^{-15}
redund_R8	2	9	7.23×10^{-15}

All eight agree, worst case 1.2×10^{-14}. The redundancy family is the more interesting half: its constraint Jacobian keeps rank 1 while its row count grows to 9, so the multipliers become underdetermined while the accelerations stay unique. The engine tracks the reference throughout. An exactly redundant constraint set does not determine how the constraint force is shared out, only its total, and neither the engine nor the reference is troubled by that.

Three points are recorded rather than corrected.

- **`loops_N4` and `loops_N5` are baseline timeouts**, not adjudicated cases; they are already counted among the ten. Given time well past the 180 s wall, `loops_N4` agrees at 7.7×10^{-15} — so its timeout is a cost limit, not an inability.
- **The suite's constrained initial conditions are rounded to four decimal places**, so every constrained case begins about 10^{-4} off its own constraint manifold. This is harmless, because the classifier measures residual *drift* from $t = 0$ rather than its absolute value, but a reader checking the residual directly would find a violation and reasonably suspect the engine.
- **The reference is deliberately left unstabilised** — no Baumgarte term, no projection — because a stabilised reference would be solving different equations from the ones the engines are asked for. The resulting drift of order 10^{-4} over three seconds is the cost of that choice, and it does not affect the derivation comparison, which evaluates accelerations at given states.

With this, **every adjudicated case that can be checked against a library-independent derivation has been.**

The coverage figures need care, because the two referees cover different sets and the increment is not additive. Counted directly from the run record over the 45 adjudicated cases:

Referee	Cases	Composition
SymPy oracle (original)	22	Lagrangian pathway only, including 5 `symbolic` cases
Library-independent reference	35	`dof` 7, `mass_ratio` 14, `redundancy` 7, `near_singular` 6, `loops` 1
Union: any independent check	**40**	
of which, gained by this work	18	
of which, only the SymPy oracle reaches	5	the `symbolic` Lagrangian cases
Still unchecked	**5**	the `symbolic` Hamiltonian cases

So 22 + 18 = 40, not 22 + 13 = 35: the reference does not cover the five `symbolic` Lagrangian cases the SymPy oracle does, and no library-independent reference for them can exist — nested cosine depth is only meaningful to an engine that consumes a raw Lagrangian, so there is no system to derive independently. The five `symbolic` Hamiltonian cases remain adjudicated by energy and structural checks alone, and that is now the whole of the study's residual blind spot.

No disagreement was found anywhere in the 40.

5.5 The single-family limitation, and the second family

This is the objection most likely to sink the paper at review. Every result in the study comes from one planar pendulum chain plus two constrained families built on the same ingredients: point masses, revolute freedom, quadratic constraints, a diagonal or trigonometric mass matrix. A reviewer will ask whether the agreement is a property of dynamics engines or a property of this particular problem. On the present evidence that question cannot be answered.

A structurally different second family would convert the work from a case study into a study. The requirement is that it exercise machinery none of the current systems touch, while remaining transcribable into all three engines without a modelling argument.

Recommended family: the slider–crank

Property	Why it matters here
Mixed joint types	A prismatic slider alongside revolute joints. Nothing in the current suite has a translational degree of freedom, so an entire class of Jacobian structure is untested.
Genuine closed loop	Crank, connecting rod, and slider form a loop whose constraint must be maintained, unlike the open chain.
Dead-centre singularities	At top and bottom dead centre the mechanism is at a *kinematic* singularity: the constraint Jacobian loses rank for geometric reasons, not numerical ones. This is a failure mode the study has not tested at all — `near_singular` degrades a mass matrix, whereas this degrades a constraint Jacobian at a specific configuration the mechanism passes through every cycle.
Portable by transcription	Drake models it natively; SymPy takes it through `LagrangesMethod` with holonomic constraints; MechanicsDSL through its constrained pathway. No modelling argument is required of any of them, unlike `near_singular` (TR-2026-03 §5).
Standard and citable	A textbook mechanism with a known closed-form solution, so the library-independent reference is tractable.

The dead-centre behaviour is the reason to prefer it over a cart-pole or a double slider. A cart-pole adds a prismatic joint but no loop and no singularity; it would test less than the chain already does. The slider–crank passes through a configuration where the mechanism's motion is genuinely indeterminate at first order, twice per revolution, in normal operation — and what an engine reports there is exactly the study's question.

Scheduling

The work is a reference implementation, three adapters, and a sweep: comparable to the whole of Week 1, which took roughly a day of concentrated effort but rested on infrastructure that now exists. The referee pattern, the adapter interfaces, the pinned-integrator comparison, and the scoring are all reusable. A realistic estimate is several days rather than several weeks, but it is not a single evening, and it competes with an academic term.
Recommendation. Do not gate the preprint on it.

1. Release the preprint as a single-family study, with the limitation stated plainly in its own section rather than buried in threats to validity. A stated limitation is a boundary; an unstated one that a reviewer discovers is a flaw.

2. Add the slider–crank afterwards. If it agrees, the null result becomes substantially stronger and generalises beyond one topology. If it disagrees — and the dead-centre singularity is the most likely place in this study for an engine to differ — then claim (B) of §5.3 becomes supported, and that is the original paper after all.

Either outcome is worth having, and neither requires delaying what is already finished.

5.6 Open decisions

Unchanged: whether the `loops` axis enters the cross-engine claim, and whether to re-express the two spring systems as pendulum chains so they become portable to a rigid-body engine (TR-2026-03 §5). Integrator pinning is now settled in practice — it is used, and §4.1 records why.

6 Artefacts and disclosure

Artefacts. `stress_suite/axis_amplitude.py`, `sweep_amplitude.py`, `compare_trajectories.py`, `out/amplitude.json`, `out/traj_chaotic.json`. Committed at `eb4ec64`.

```
# inside WSL, where all three engines import in one process
PYTHONPATH=<repo>/src ~/drake-venv/bin/python sweep_amplitude.py
PYTHONPATH=<repo>/src ~/drake-venv/bin/python compare_trajectories.py --amp 3.0
```

Disclosure. The author is the maintainer of MechanicsDSL, one of the three engines measured. This is disclosed in every document of the study. The mitigation is structural: no engine adjudicates itself or any other, and the referee shares no library with any of them. AI assistance was used in the derivation, implementation, measurement, and preparation of this report. Every figure was produced by executing the committed code.